

Type 2 Diabetes Risk Prediction

A comparative analysis of predictive modelling techniques with a focus on calibration and explainability

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ROADMAP

Agenda

- 1 Introduction
- 2 Literature Review
- 3 Data and Methodology
- 4 Results
- 5 Discussion and Conclusions
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1

Introduction

1 Diabetes burden and gaps in the literature

GLOBAL AND SOUTH AFRICAN CONTEXT

- Diabetes accounts for **5.9%** of all deaths in South Africa (2023), and **7.5%** of female deaths [1].
- Globally, **537 million** adults lived with diabetes in 2021; projected to reach **783 million** by 2045 [2].
- Lifestyle interventions for prediabetic patients can delay or prevent progression to Type 2 diabetes.

Two recurring gaps

- Discrimination reported, calibration ignored [3].
- Explainability reported for a single model, no cross-model comparison [8].

This study

A structured four-model comparison on the CDC BRFSS 2015 dataset, jointly evaluating [discrimination](#), [calibration](#), and [explainability](#), with decision-curve analysis for clinical utility.

1 Research objectives

1. Review existing predictive modelling techniques and their applications in public health.
2. Collect and preprocess a public health dataset from a publicly available surveillance system.
3. Develop predictive models using statistical and machine learning techniques for Type 2 diabetes risk.
4. Compare model performance using **discrimination, calibration, and validation** metrics.
5. Identify key risk factors and assess their **consistency across models**.
6. Propose a data-driven decision-support framework informed by calibration and explainability analyses.

2

Literature Review

- Approximately **one in nine adults** lived with diabetes globally in 2021; many cases in low- and middle-income countries remain undiagnosed [2].
- In South Africa, diabetes is now a leading contributor to premature mortality [1].
- The CDC's **BRFSS** is the largest telephone-based behavioural risk factor survey, supporting diabetes risk research for over two decades.
- The 2015 BRFSS cohort is a public benchmark for binary diabetes classification.
- Comparative BRFSS studies apply logistic regression alongside tree-based ensembles; accuracy in the **0.80–0.86** range.
- If models rank patients similarly, on what basis should one be preferred?

Four model families

- **Elastic Net**—penalised linear [4]
- **GAM**—additive, interpretable [5]
- **Random Forest**—bagging [6]
- **XGBoost**—boosting [7]

Spanning the penalised, additive, bagging, and boosting traditions.

2 Calibration, explainability, and clinical utility gaps

Calibration is underdeveloped

- Discrimination dominates but calibration rarely reported [3].
- Standard tools: Brier score, ECE, Hosmer–Lemeshow, reliability diagrams.
- BRFS studies rarely apply post-hoc recalibration (Platt, isotonic).
- SMOTE distorts the joint distribution and damages calibration.

Explainability and DCA gaps

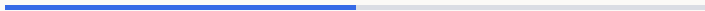
- SHAP [8] typically applied to a single model.
- Cross-model SHAP comparisons and ICE curves are scarce.
- Decision curve analysis [9]—quantifying clinical utility via net benefit—is essentially absent from the BRFS diabetes literature.

Contribution

A four-model comparison with cross-family SHAP, calibration, class weighting over SMOTE, and decision-curve analysis—exposing trade-offs that single-model studies miss.

3

Data and Methodology



3 Data source and variables

CDC BRFSS 2015

- **253 680 respondents**, 22 columns (21 predictors + outcome).
- Imbalanced binary file is the primary dataset—preserves prevalence for calibration.
- Outcome: 86.1% no diabetes, 13.9% diabetes ($\approx 6.2:1$ imbalance).
- 21 predictors across demographic, clinical, behavioural, and socioeconomic groups.

Table 1. Selected variables. Binary as $n(\%)$; others as mean (SD), median.

Variable	Type	Summary
Diabetes_binary	Binary	13.9% positive
Sex	Binary	44.0% male
Age	Ordinal	8.03 (3.05), 8
Education	Ordinal	5.05 (0.99), 5
Income	Ordinal	6.05 (2.07), 7
HighBP	Binary	42.9%
HighChol	Binary	42.4%
BMI	Continuous	28.4 (6.61), 27
GenHlth	Ordinal	2.51 (1.07), 2
MentHlth	Count	3.18 (7.41), 0
PhysHlth	Count	4.24 (8.72), 0
Smoker	Binary	44.3%
PhysActivity	Binary	75.6%

3 Exploratory findings and preprocessing

EDA highlights

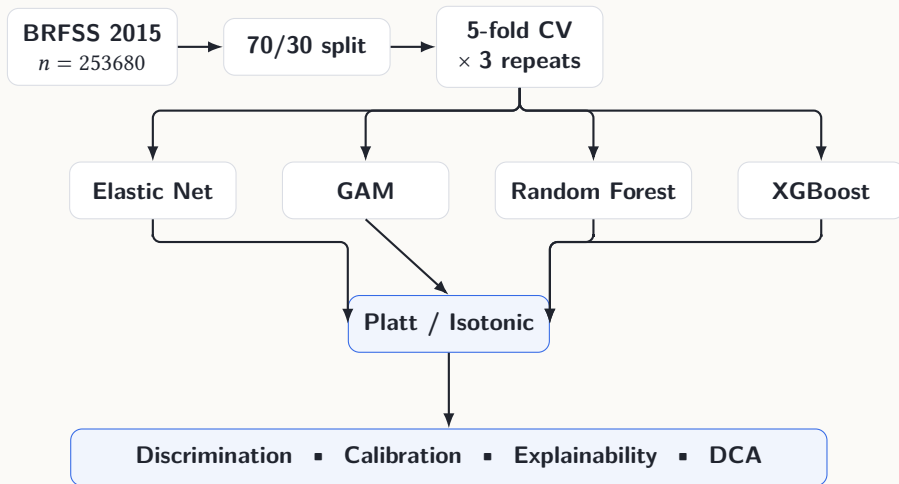
- Mean BMI rises from 27.7 (no diabetes) to 31.9 (diabetes), with substantial overlap.
- Strongest correlates: GenHlth ($\rho = 0.30$), HighBP (0.27), BMI (0.24); Income -0.17 .
- HighBP has the largest Cramér's V (0.263); AnyHealthcare the smallest (0.016).
- Prevalence rises monotonically with age, falls with income.

Preprocessing decisions

- Binary $\rightarrow$ factors; ordinal $\rightarrow$ ordered factors; BMI/MentHlth/PhysHlth numeric.
- No missing values in the released file (CDC-imputed).
- BMI outliers (up to 98) retained; Elastic Net winsorisation sensitivity check.
- **Class weighting** ($w_0=0.162$, $w_1=0.838$) instead of SMOTE—preserves calibration.
- 70/30 stratified split; repeated 5-fold CV ($\times 3$) [10].

3

Analytical pipeline



Four model families are tuned under a common `tidymodels` workflow, then post-hoc calibrated and evaluated along four complementary axes on the held-out test set.

3 Models and evaluation

Four model families

- **Elastic Net** [4]: $\ell_1 + \ell_2$ penalised logistic regression; linear, interpretable baseline with implicit variable selection.
- **GAM** [5]: penalised regression splines for continuous/ordinal predictors; parametric but non-linear; well calibrated by construction.
- **Random Forest** [6]: bagged trees with class-weighted bootstrap.
- **XGBoost** [7]: boosted regression trees with early stopping on the logistic loss.

Evaluation

- **Discrimination:** AUC, AUPRC, Youden-optimal sens./spec.; pairwise DeLong tests with Bonferroni correction.
- **Calibration:** Brier score, ECE, reliability slope, Hosmer–Lemeshow; Platt scaling vs isotonic regression.
- **Explainability:** TreeSHAP for RF/XGBoost, KernelSHAP for Elastic Net/GAM [8], permutation importance, PDPs and ICE curves.
- **Clinical utility:** decision curve analysis [9] over $t \in [0.05, 0.40]$.

4

Results



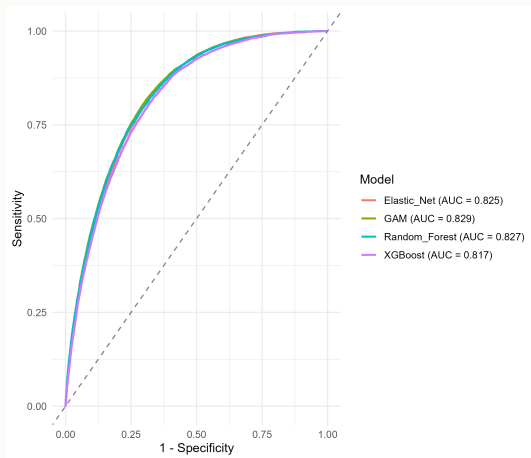


Figure 1. ROC curves on the held-out test set ($n = 76105$).

Table 2. Test-set discrimination metrics.

Model	AUC	AUPRC	Sens.	Brier
Elastic Net	0.825	0.738	0.791	0.0988
GAM	0.829	0.734	0.815	0.0970
Random Forest	0.827	0.735	0.793	0.0973
XGBoost	0.817	0.740	0.807	0.1002

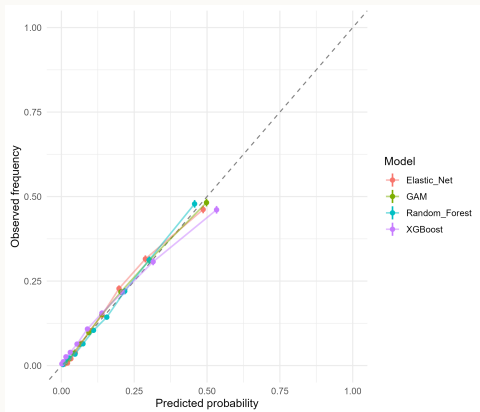


Figure 2. Reliability diagrams for raw (uncalibrated) predictions.

Table 3. ECE by model and recalibration method (lower is better).

Model	Raw	Platt	Isotonic
Elastic Net	0.0143	0.0148	0.0040
GAM	0.0073	0.0078	0.0028
Random Forest	0.0086	0.0260	0.0274
XGBoost	0.0158	0.0066	0.0044

GAM well calibrated out of the box; Elastic Net and XGBoost benefit from isotonic regression; **Random Forest is the only model degraded by recalibration**—challenging the idea that tree ensembles should be recalibrated by default.

4 Explainability

Table 4. Permutation importance (drop in AUC). Strong cross-model agreement on the top five features.

	Feature	EN	GAM	RF	XGB
1	GenHlth	0.064	0.066	0.044	0.060
2	BMI	0.027	0.034	0.032	0.036
3	Age	0.022	0.034	0.024	0.033
4	HighBP	0.024	0.018	0.023	0.022
5	HighChol	0.012	0.010	0.010	0.009

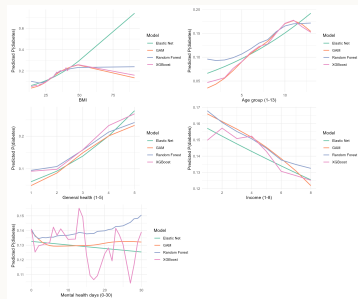


Figure 3. Partial dependence plots for the top continuous predictors.

XGBoost's ranking diverges from permutation importance—overweighting rare binary features (Stroke, Smoker) with large marginal log-odds but small AUC contribution. PDPs show monotonic BMI and Age effects with a steeper gradient above BMI 30; BMI contribution is amplified for older individuals.

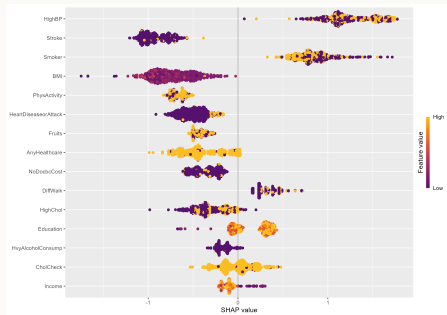


Figure 4. XGBoost SHAP beeswarm plot for the top 15 features. Each point is one observation; x-axis is the SHAP value in log-odds units; points coloured by feature value.

- GenHlth, HighBP, BMI have the widest spreads.
- BMI SHAP values coloured by Age: positive BMI contribution is **amplified for older individuals**.
- High feature values (red) push risk up for GenHlth, BMI, HighBP; down for Income, PhysActivity.

4 Subgroup performance and clinical utility

Subgroup analysis

- GAM has the highest AUC in 5 of 7 subgroups and the lowest ECE in 6 of 7.
- All four models discriminate worse in 65+ and low-income subgroups (case-mix effect).
- XGBoost's calibration degrades sharply in the low-income subgroup (ECE 0.037).
- GAM maintains ECE < 0.015 across all subgroups.

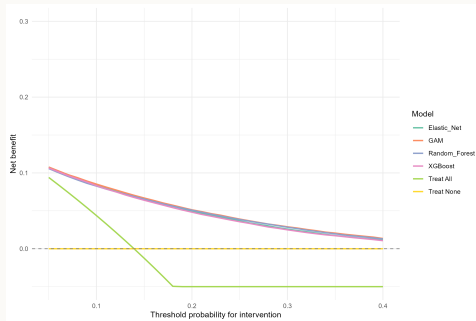


Figure 5. Decision curves: net benefit vs. threshold.

All four models dominate treat-all above $t \approx 0.13$; treat-all is harmful above $t \approx 0.15$. At $t = 0.20$, all models still deliver positive net benefit (0.048–0.051) while treat-all is at -0.076 .

5

Discussion and Conclusions



5 Principal findings

Central result

The four models were **practically equivalent in discrimination** (AUC 0.817–0.829) but **differed substantially in calibration behaviour** and in their response to recalibration.

- **GAM** well calibrated without adjustment—additive models fit the conditional expectation directly.
- **Elastic Net** benefited modestly from isotonic regression; **XGBoost** benefited substantially.
- **Random Forest** was the *only* model degraded by recalibration—recalibration should be evaluated empirically, not assumed.
- Cross-model SHAP agreement on GenHlth, BMI, Age, HighBP, HighChol is reassuring for clinical use.
- GAM was simultaneously the best-discriminating, the best-calibrated, and among the best in net benefit.

Limitations

Self-reported data; calibration fitted on a second split (slightly optimistic); single US dataset; over-powered DeLong/HL tests at $n = 76105$.

5 Conclusions and future work

What this study adds

A comparative, calibration-aware methodology on a well-characterised surveillance dataset exposes trade-offs that single-model, discrimination-focused studies miss.

- **Calibration** should be reported alongside discrimination whenever predicted probabilities drive clinical decisions.
- **Recalibration is not universally beneficial**—each model family's response must be evaluated empirically.
- **Class weighting** over SMOTE preserves calibration of predicted probabilities.
- **Cross-family SHAP** comparisons and **decision curve analysis** should be standard in diabetes risk prediction research.
- **Future work:** external validation on a South African cohort; subgroup-specific calibration; cost-sensitive decision curves with asymmetric false-positive/false-negative costs.

6

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Thank you

Questions and discussion are welcome.

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